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Photobiomodulation for Managing "Dry Socket": A Randomised Controlled Trial

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ABSTRACT

Objective: There are no studies that have evaluated the effect of Alveogyl with and without adjunct photobiomodulation therapy (PBMT) in reducing self-rated post-operative pain (SPP) in patients with alveolar osteitis (AO; dry socket). The aim of the present randomised controlled trial was to assess the effectiveness of Alveogyl with and without PBMT for the management of SPP in patients with AO.

Methods: Adult nonsmokers with diagnosed AO were included. Patients were randomly divided into 4 groups. In Group 1, patients underwent mechanical curettage (MC) with copious normal saline irrigation. In Group 2, patients underwent MC + Alveogyl dressings in extraction sites which were changed every 48 hours until cessation of pain. In Group 3, patients underwent MC + Alveogyl followed by PBMT using a 660-nm indium-gallium-aluminum-phosphide diode laser. In Group 4, patients were treated solely with PBMT. The visual analogue scale was used up to 3 postoperative days to assess SPP up to 3 days at 6- (T0) and 12-hour (T1) intervals. Statistical analyses were performed using the analysis of variance and Bonferroni post hoc adjustment tests. Correlation between SPP scores and age, sex, and eruption status of teeth was assessed using logistic regression models. *P* values <.01 were nominated as being statistically significant.

Results: In all, 14, 13, 14, and 14 individuals with AO were included in groups 1, 2, 3 and 4, respectively. All patients had undergone extraction of mandibular third molars. At baseline and on day 1, there was no difference in SPP in all groups. On days 2 and 3, mean visual analogue scale (VAS) scores at T1 ($P < .01$) and T2 ($P < .01$) intervals were significantly high in Group 2 compared with Group 3. On days 2 and 3, mean VAS scores at T1 ($P < .01$) and T2 ($P < .01$) intervals were significantly high in Group 4 compared with Group 3. There was no difference in SPP in groups 3 and 4 on day 3 at T0 and T1 intervals.

Conclusions: PBMT following MC and Alveogyl dressing is more efficient in reducing SPP compared with MC with or without Alveogyl dressing in patients with AO.

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Introduction

Alveolar osteitis (AO), commonly referred to as "dry socket," is a complication that is often encountered after exodontia[1] and occurs between the first and the third postoperative days

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due to partial or complete fibrinolysis at the extraction site. Clinically, AO is manifested as excruciating pain at the extraction site that often radiates to the ophthalmic, temporal, auricular, and/or cervical regions.[1] Other symptoms of AO include halitosis, limited mouth opening, underlying bone exposure covered by a gray-yellow layer of necrotic tissue, and impaired mastication.[1-3] Although AO occurs in up to 5% of tooth extractions[4]; it is encountered in 30% of mandibular third molar (MTM) extractions.[5] Common risk factors for AO include traumatic extraction, habitual use of tobacco products and oral contraceptives, female sex, and preexisting infections such as chronic apical periodontitis at sites of extraction.[1,6-8]

There is a debate in indexed literature regarding the most precise therapeutic protocol for the management of AO. Traditionally, a paste containing a mixture of zinc oxide and eugenol (ZOE) is used for the treatment of AO.[9] Other pharmacologic preparations including 0.2% chlorhexidine, hyaluronic and aminocaproic acids, and platelet-rich fibrin have been proposed to significantly help in reducing postoperative pain in patients with AO[10-13]; however, a mixture of 25.7 grams of butamben, 15.8 grams of iodoform, and 13.7 grams of eugenol (Alveogyl, Septodont) has attracted sufficient attention by clinicians and researchers in this regard.[9,14,15] According to Chaurasia et al,[9] dressings of ZOE paste are more effective in minimising self-rated postoperative pain (SPP; measured using the visual analogue scale [VAS]) than Alveogyl amongst patients with AO. Photobiomodulation therapy (PBMT) has emerged as a promising and effective treatment strategy for the management of orodental and craniofacial disorders as it exhibits anti-inflammatory, antinociceptive, and antimicrobial effects.[16-21] Randomised controlled trials (RCTs)[22-24] have investigated the influence of PBMT in reducing SPP in patients with AO following extraction of MTM. According to Jovanović et al,[23] PBMT once daily for 8 days in extraction sockets with AO results in a statically significant reduction in SPP compared with extraction sockets that are treated with ZOE. Although the effectiveness of Alveogyl in the treatment of AO cannot be overlooked, there are no studies that have evaluated the effect of PBMT with and without adjunct PBMT in reducing SPP in patients with AO.

The aim of the present RCT was to assess the effectiveness of Alveogyl with and without PBMT for the management of SPP in patients with AO. The null hypothesis was that there is no difference in SPP when Alveogyl dressings are applied in extraction sockets with or without adjunct PBMT.

Materials and methods

Study design

The present study is an RCT registered with Sharavathi Dental College and Hospital, Shivamogga, Karnataka, 577204, India (Registration # SDC/SMG/2022/089).

Ethics declaration

The study protocol was reviewed and approved by the Institutional Review Board of the Sharavathi Dental College and Hospital, Shivamogga, Karnataka, 577204, India (04/20212/CR). An information sheet was provided to all individuals that provided the purpose and methodology of the present trial was provided to all participants. Individuals who agreed to participate in the present trial were requested to sign a consent form. All individuals were aware that participation is completely voluntary and withdrawal at any stage would not lead to penalties or/and consequences. All individuals regardless of the decision to volunteer or withdraw were invited to ask questions.

Study location

The present study was performed at the Department of Oral Medicine and Radiology, Sharavathi Dental College and Hospital, Shivamogga, Karnataka, 577204, India. The study was performed between October 2021 and March 2022.

Patient eligibility protocol

Adult patients diagnosed with AO were included. Diagnosis of AO was based on characteristics described by Blum,[3] which are as follows: (a) partial or total loss of blood clot within the extraction socket with or without halitosis and (b) SPP within and around the extraction socket that increases in severity at any time between the first and third day after the extraction. Pregnant and/or lactating females and self-reported tobacco smokers and smokeless tobacco product users were excluded. All individuals who agreed to participate in the present study were mandated to read and sign a written informed consent form.

Questionnaire

The following data were gathered using a questionnaire: (a) age in years; (b) sex (male, female, or prefer not to disclose), (c) tobacco smoking (yes or no), (d) use of smokeless tobacco products (yes or no), (e) systemic diseases (yes or no), and (f) pregnancy and/or lactation (yes or no).

Evaluation of dental records

Patients' dental records were evaluated to determine (a) the tooth/teeth socket/sockets associated with AO; (b) the eruption status of the affected tooth/teeth (fully erupted, partially erupted, impacted [soft tissue or bony]); (c) angulation of the impacted tooth/teeth (horizontal, mesioangular, distoangular, or vertical); (d) jaw location of tooth/teeth associated with AO; (e) surgical procedure (such as elevation of flaps, osteotomy, suture placement); and (f) duration of procedure in minutes (from incision to suturing).

Randomisation, study groups, and allocation concealment

Randomisation was done by picking a paper marked "Group 1," "Group 2," "Group 3," or "Group 4" from an opaque box. All patients underwent mechanical curettage (MC) of extraction site with copious normal saline (NS) irrigation under local anesthesia (Lidocaine 2%, Epinephrine 1:100 000). Therapeutically, participants were divided into 4 groups. In Group 1,

patients underwent MC with copious NS irrigation. In Group 2, MC was done and Alveogyl dressing was placed in extraction sites and changed every 48 hours until cessation of pain. In Group 3, MC was done and Alveogyl dressing was placed in extraction sites followed by PBMT. In Group 4, patients were treated solely with PBMT.

Photobiomodulation

The protocol described in the study by Eshghpour et al[22] and Kamal et al[25] was used for PBMT. In summary, PBMT was done following adoption of laser safety protection measures such as wearing of goggles by patient and operator and a warning sign installed outside the operatory which stated "Caution! Laser in use." PBMT was done using a 660-nm indium-gallium-aluminum-phosphide (InGaAlP) diode laser (Thor-DD2-control unit, Thor). The laser was used in continuous wave mode and at a power and energy fluence of 200 milliwatts (mW) and 85.7 joules per square centimeter (cm^2), respectively. The surface area of the probe used for laser delivery was 0.07 cm^2. The following surfaces were irradiated: (a) middle of the socket and (b) lingual and buccal surfaces of the extraction socket. Each of these surfaces were irradiated for 30 seconds and each surface was delivered 6 joules of energy. The laser tip was kept at an approximate delivery distance of 1 centimeter from the surface of the extraction socket.

Assessment of self-rated postoperative pain

The VAS,[26] which consists of a 10-cm horizontal scale with "0" indicating no pain and "10" demarcating worst imaginable pain was used for quantification of SPP in all groups. Patients were requested to record scores for SPP immediately before intervention/baseline (T0) and 6 (T1) and 12 (T2) hours later for 3 days. Patients who continued to perceive pain after 1 week of treatment were scheduled for additional follow-up visits.

Sample-size estimation and statistical analyses

Sample-size estimation was done on data from a pilot investigation. It was estimated that at least 13 individuals would be needed in each group in order to detect a 2-cm difference in VAS with an alpha of 1%. With this sample size, the study was estimated to have a power of 80%. Commercial software (IBM, SPSS Version 22) was used to perform quantitative analyses. Data normality was determined using the Kolmogorov–Smirnov test. One-way analysis of variance and Bonferroni post hoc adjustment tests were done to compare SPP amongst the groups. Correlation between SPP and age, sex, and eruption status of teeth whose extraction sockets demonstrated AO were assessed using logistic regression models. *P* values below .01 were considered statistically significant. The statistician was blinded to the study groups.

Results

Participants and grouping

In all, 14, 13, 14, and 14 individuals with AO were included in groups 1, 2, 3, and 4, respectively. In groups 1, 2, 3 and 4, 10, 10, 11, and 10 patients were female. There was no significant difference in the mean age of patients in groups 1 (34.3 ± 2.1 years), 2 (32.9 ± 0.5 years), 3 (33.1 ± 1.2 years), and 4 (32.6 ± 1.4 years). All patients had AO in extraction sites of mandibular third molars. Of these, 11, 10, 8, and 10 teeth were fully erupted in groups 1, 2, 3, and 4, respectively (Table 1). All patients had been prescribed postoperative antibiotics (amoxicillin 500 mg) and nonsteroidal anti-inflammatory drugs (ibuprofen 800 mg) by their oral health care providers (OHPs) who had performed extractions of MTM.

Self-rated post-operative pain

- *Baseline*

At baseline, there was no significant difference in the mean SPP amongst patients in groups 1 (8.78 ± 0.3), 2 (8.4 ± 0.2), 3 (8.73 ± 0.4), and 4 (8.77 ± 0.3), respectively.

- *Intragroup comparisons*

In Group 1, there was no significant difference in the mean SPP when T0 VAS scores were compared with T1 and T2 up to 3 days. There was a statistically significant reduction in the mean VAS scores compared with baseline in Group 2 ($P < .01$) on day 3. In Group 3, there was a statistically significant reduction in the mean VAS scores at T1 and T2 intervals on days 2 ($P < .01$) and 3 ($P < .01$). In Group 4, there was a statistically significant reduction in the mean VAS scores at T1 and T2 intervals on day 3 ($P < .01$; Table 2).

Table 1 – Characteristics of the study groups.

Parameters	Group 1	Group 2	Group 3	Group 4
Number of patients (n)	14	13	14	14
Mean age	34.3 ± 2.1 years	32.9 ± 0.5 years	33.1 ± 1.2 years	32.6 ± 1.4 years
Sex	4 male	3 male	3 male	4 male
	10 female	10 female	11 female	10 female
Tooth/teeth associated with AO, No.				
Mandibular right third molar	5	7	10	12
Mandibular left third molar	9	6	4	2
Mandibular right and left third molar	—	—	—	—
Eruption status, No.				
Fully erupted	11	10	8	10
Partially erupted	3	3	5	4

AO, alveolar osteitis.

Table 2 – Self-rated pre and postoperative pain scores.

Study groups	Baseline	Day 1		Day 2		Day 3	
	T0	T1	T2	T1	T2	T1	T2
Group 1	8.78 ± 0.3	8.5 ± 0.2	8.52 ± 0.3	8.21 ± 0.4	8.07 ± 0.3	7.71 ± 0.2	7.43 ± 0.2
Group 2	8.44 ± 0.2*	7.07 ± 0.08	6.43 ± 0.2	4.21 ± 0.3	4 ± 0.3	2.57 ± 0.08	2.07 ± 0.06
Group 3	8.73 ± 0.4†	7.42 ± 0.2	6.93 ± 0.2	2 ± 0.05	2.21 ± 0.03	0.64 ± 0.04	0.57 ± 0.01
Group 4	8.77 ± 0.3	8.29 ± 0.4	7.57 ± 0.6	5.86 ± 0.3	5.36 ± 0.4	1.35 ± 0.07	1.14 ± 0.04

T0: Baseline. T1: After 6 hours. T2: After 12 hours.

* Compared with T1 ($P < .01$) and T2 ($P < .01$) on day 3.
† Compared with T1 and T2 on days 2 ($P < .01$) and 3 ($P < .01$).
‡Compared with T1 ($P < .01$) and T2 ($P < .01$) on day 3.
Group 1: Mechanical curettage with copious normal saline irrigation.
Group 2: After mechanical curettage, Alveogyl (Septodont) dressing was placed in extraction sites and changed every 48 hours until cessation of pain.
Group 3: After mechanical curettage, Alveogyl dressing was placed in extraction sites followed by photobiomodulation therapy (PBMT).
Group 4: PBMT alone.

- *Intergroup comparisons*

Group 1 vs Group 2

At baseline and on day 1, there was no significant difference in mean VAS score amongst individuals in groups 1 and 2. On days 2 and 3, mean VAS scores at T1 ($P < .01$) and T2 ($P < .01$) intervals were significantly high in Group 1 compared with Group 2 (Figure 1).

Group 1 vs Group 3

At baseline and on day 1, there was no significant difference in mean VAS score amongst individuals in groups 1 and 3. On days 2 and 3, mean VAS scores at T1 ($P < .01$) and T2 ($P < .01$) intervals were significantly high in Group 1 compared with Group 2 (Figure 2).

Group 1 vs Group 4

At baseline and on day 1, there was no significant difference in mean VAS score amongst individuals in groups 1 and 3. On days 2 and 3, mean VAS scores at T1 ($P < .01$) and T2 ($P < .01$) intervals were significantly high in Group 1 compared with Group 2 (Figure 3).

Group 2 vs Group 3

At baseline and on day 1, there was no significant difference in mean VAS score amongst individuals in groups 2 and 3. On days 2 and 3, mean VAS scores at T1 ($P < .01$) and T2 ($P < .01$) intervals were significantly high in Group 2 compared with Group 3 (Figure 4).

Group 2 vs Group 4

There was no statistically significant difference in mean VAS scores up to 3 days of follow-up (Figure 5).

Group 3 vs Group 4

At baseline and on days 1 and 3, there was no significant difference in mean VAS score amongst individuals in groups 3 and 4. On days 2 and 3, mean VAS scores at T1 ($P < .01$) and T2 ($P < .01$) intervals were significantly high in Group 4 compared with Group 3 (Figure 6).

Correlation amongst self-rated pain and age, sex, and eruption status of teeth

There was no statistically significant correlation amongst age, sex, and eruption status of teeth amongst those whose extraction sockets demonstrated AO (data not shown).

Discussion

"Dry socket" or AO is associated with agonising postsurgical pain that occurs in up to 10% of patients after exodontia.[27,28] Besides resulting in intense discomfort, AO may induce negative perceptions amongst patients such as financial burdens, dental anxiety, and fear towards OHPs and/or future dental treatments.[14,28] Alveogyl is a eugenol-based dressing that is commonly used for the management of SPP and facilitates healing in patients with AO.[14] The authors of the present study therefore hypothesised that treatment of AO using Alveogyl in combination with PBMT (Group 3) is more effective in reducing SPP compared with other treatment protocols

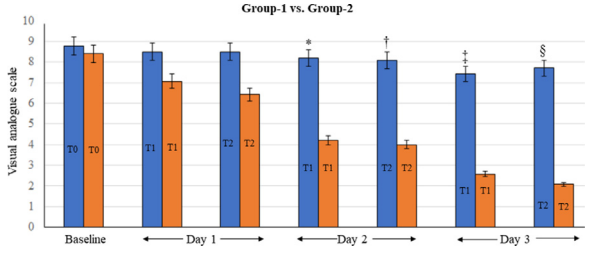


Fig. 1 – Mean self-rated postoperative pain scores at T1 and T2 intervals in groups 1 (blue bars) and 2 (red bars). *Compared with T1 in Group 2 ($P < .01$). †Compared with T2 in Group 2 ($P < .01$). ‡Compared with T1 in Group 2 ($P < .01$). §Compared with T2 in Group 2 ($P < .01$).

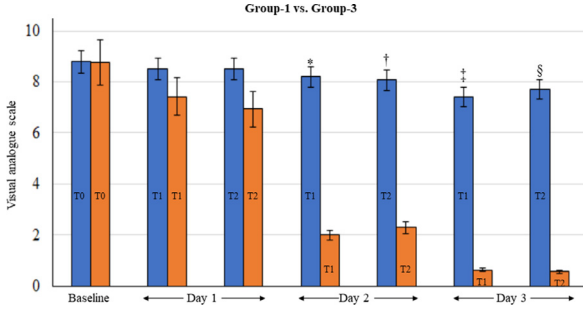


Fig. 2 – Mean self-rated postoperative pain scores at T1 and T2 intervals in groups 1 (blue bars) and 3 (red bars). *Compared with T1 in Group 3 ($P < .01$). †Compared with T2 in Group 3 ($P < .01$). ‡Compared with T1 in Group 3 ($P < .01$). §Compared with T2 in Group 3 ($P < .01$).

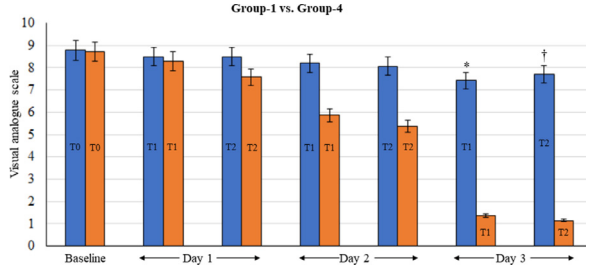


Fig. 3 – Mean self-rated postoperative pain scores at T1 and T2 intervals in groups 1 (blue bars) and 4 (red bars). *Compared with T1 in Group 4 ($P < .01$). †Compared with T2 in Group 4 ($P < .01$).

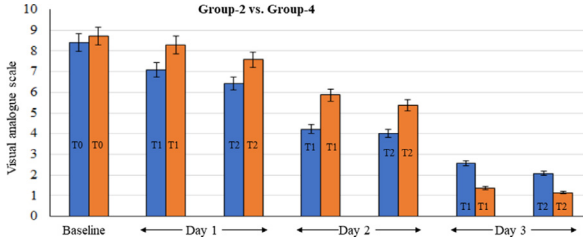


Fig. 5 – Mean self-rated postoperative pain scores at T1 and T2 intervals in groups 2 (blue bars) and 4 (red bars).

such as MC alone (Group 1), MC + Alveogyl dressing (Group 2), and PBMT alone (Group 4). Our results showed a statistically significant reduction in SPP at days 2 and 3 compared with baseline amongst patient groups 1 and 2. Our results, as demonstrated in Figure 1, are in contradiction with those reported in the study by Chaurasia et al,[9] which found that ZOE is more effective in reducing SPP than is Alveogyl. It is suggested that the results reported in the study by Chaurasia et al[9] should be cautiously interpreted, as the sample size was not power-adjusted. Therefore, the possibility of *P* values being biased cannot be ruled out. With regards to PBMT, studies[14,22] have shown that low-level laser therapy is a useful therapeutic regimen for reducing SPP in patients with AO. To our knowledge from pertinent indexed literature, the present study is the first one to compare the effects of Alveogyl with and without adjunct PBMT on SPP in patients with AO. According to Khalighi et al,[29] PBMT when performed with low-level lasers at an output power of less than 250 mW induces photobiochemical reactions that reduce tissue levels of pain mediators such as serotonin, acetylcholine, and histamine and facilitates healing without affecting tissue temperatures. The present results showed a statistically significant reduction in SPP at day 2 amongst patients who underwent MC + Alveogyl dressings + PBMT (Group 3) compared to patients who received MC + Alveogyl dressing (Group 2) alone (Figure 4). Moreover, our results showed that by day 3, the SPP scores were markedly low and ranged between 0 and 2 amongst most of the patients in groups 3 (mean VAS score ~0.6) and 4 (mean VAS score ~1). These results reflect that PBMT after application of Alveogyl dressings is more effective in reducing AO-associated SPP than when PBMT or Alveogyl are used as sole treatment strategies following MC. Though it is difficult to offer precise reasons for the foregoing observations, the following conclusions could be drawn from the available data.[30] According to Chen et al,[30] PBMT increases the expression of prostatic acid phosphatase (an analgesic mediator) in tissues. It has also been proposed that PBMT reduces pain perception by increasing production of serotonin, reducing the velocity of action potential (AP) conduction, and increasing latency in sural and median nerves.[31] Similarly, eugenol-based dressings such as Alveogyl interrupt AP by stabilising neuronal membranes and inhibiting sodium currents in dental afferent neurons.[32] It is tempting to speculate that PBMT when performed in conjunction with Alveogyl dressings further augments antinociceptive efficacy by enhancing serotonin production and decreasing AP in dental nerves in contrast to when these therapeutic strategies are performed independently. Further studies are needed to test this hypothesis.

Despite that Alveogyl dressings are often used for the management of AO, there are other potential therapies that have been shown to be useful in promoting healing and reducing SPP in patients with AO. For instance, Kamal et al[33] reported that application of concentrated growth factor dressings after MC of extraction sockets is more effective in reducing SPP and promotes healing in patients with AO compared with MC alone and saline irrigation. Results by Suchánek et al[34] showed that hyaluronic acid–based medications are useful for reducing SPP in patients with AO, whereas 0.2% chlorhexidine was shown to significantly reduce

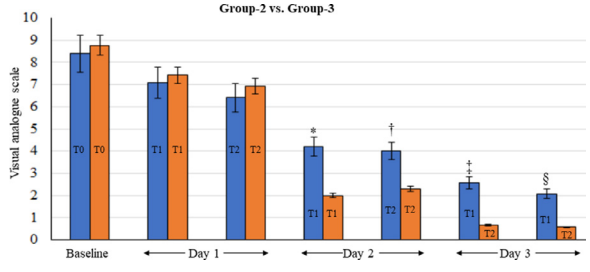


Fig. 4 – Mean self-rated postoperative pain scores at T1 and T2 intervals in groups 2 (blue bars) and 3 (red bars). *Compared with T1 in Group 3 ($P < .01$). †Compared with T2 in Group 3 ($P < .01$). ‡Compared with T1 in Group 3 ($P < .01$). §Compared with T2 in Group 3 ($P < .01$).

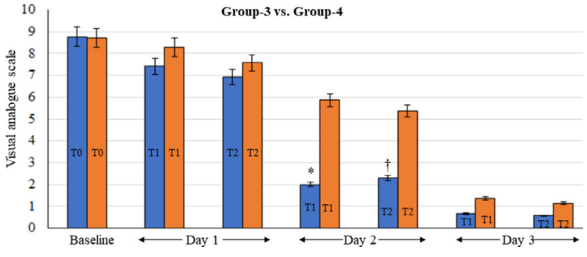


Fig. 6 – Mean self-rated post-operative pain scores at T1 and T2 intervals in groups 3 (blue bars) and 4 (red bars). *Compared with T1 in Group 4 ($P < .01$). †Compared with T2 in Group 4 ($P < .01$).

complications following extraction of MTM in the RCT by Muñoz-Cámara et al.[35] Furthermore, in a recent systematic review and meta-analysis, Zhu et al[36] reported that placement of rich fibrin dressings is a reliable treatment protocol for reducing SPP following extraction of MTM. The authors of the present study speculate that adjuvant application of PBMT with any of the aforementioned protocols would help further in reducing SPP after MTM compared to when these regimens are used alone after MC. There are no studies that have tested this hypothesis, and it warrants additional clinical trials. With regards to PBMT parameters, the authors referred to parameters that were used in previous investigations.[22,25] Moreover, PBMT was performed once in the present investigation. It is hypothesised that multiple sessions of PBMT at varying time intervals (for instance on day 2 and/or day 3) following MC and Alveogyl placement using the current approach is more efficient in reducing SPP than a single session of PBMT. Nejat et al[20] further showed that application of PBMT within a week of extraction of MTM reduces the incidence of AO and related postoperative pain. This suggests that inclusion of postsurgical PBMT following exodontia in oral surgical protocols can help minimise the risk of postsurgical complications including AP and SPP. This suggestion warrants additional studies.

Tobacco-smoking is a risk factor for AO.[6] A limitation of the present study is that tobacco smokers were excluded. It is speculated that tobacco-smoking after MC and Alveogyl placement with or without PBMT aggravates SPP in patients with AO. Because the present study had a clinical design, it was challenging to scientifically justify postoperative healing at the extraction sites as it would require histologic evaluation of target tissues. Another limitation is that microbial assessment was not performed in the present study. It has been reported that bacteria—*Peptostreptococci*, *Parvimonas*, *Fusobacterium*, and *Prevotella* species—are commonly identified in the extraction sockets of patients with AO.[37] There is a possibility that Alveogyl dressing with adjunct PBMT is more effective in reducing counts of pathogenic microbes in extraction sockets with AO compared with sites that are treated solely with Alveogyl dressing. Furthermore, based on the study design and acquired ethical approvals, additional dressing treatments such as ZOE and platelet rich fibrin with or without PBMT could not be assessed in the present investigation. Further studies are needed to test the aforementioned hypotheses.

Within the limits of the present study, it is concluded that PBMT following MC and Alveogyl dressing is more efficient in reducing SPP compared to MC with or without Alveogyl dressing in patients with AO.

Author contributions

SSA designed the study and wrote the manuscript; DA wrote the manuscript and revised it prior to submission; NZA performed the statistical analysis and wrote the results; DDD performed photobiomodulation and wrote the manuscript; MKA measured self-rated pain and wrote the discussion; and MSB wrote the manuscript and revised it prior to submission.

Conflict of interest

None disclosed.

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